Mandarin Phonetics Representations Learning via BiLSTM

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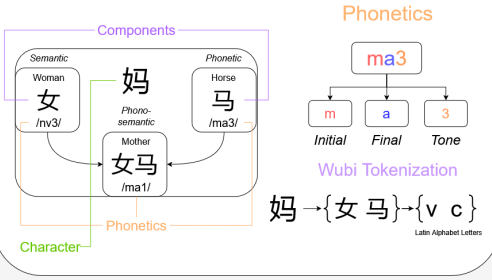
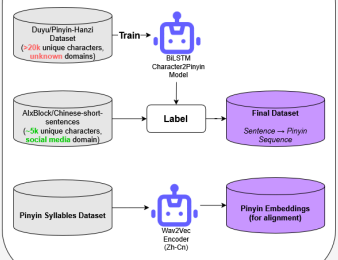
Motivation

In many representation learning tasks, researchers aim to extract embeddings of objects that reflect similarity along a target feature, such as phonetic, semantic, or visual attributes, rather than merely optimizing a discriminative loss. However, standard supervised training, like cross-entropy on discrete labels, offers no direct pressure to shape the internal geometric structure of the hidden space according to that feature's continuous similarity. To address this, I present a case study in the NLP subfield of Mandarin phonetics modeling, experimenting with several explicit strategies that enforce target-based grouping with BiLSTM encoder. These strategies include a shared input basis (radical decomposition), a continuous acoustic target space (wav2vec2 embeddings), and dimensionality reduction to reveal latent structure. Furthermore, I design an evaluation framework that decouples the influence of different factors (visual, phonetic, acoustic, semantic) on hidden-space similarity, using both regression analysis and decoder dynamics as diagnostics.

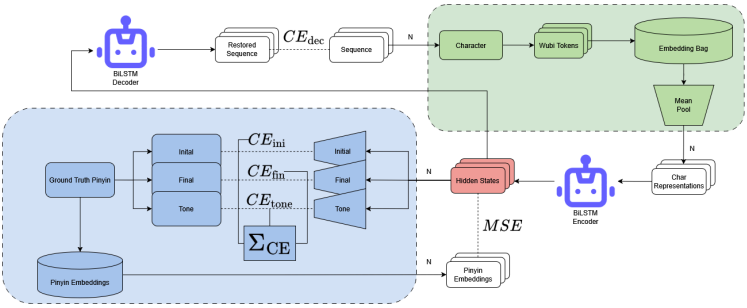
Experimental Setting



Dataset Collection



Training Pipeline



Loss Functions

$$\mathscr{L} = \text{CE}_{\text{dec}} + \boldsymbol{\lambda}_{\text{enc}} \Sigma_{\text{CE}}$$

$$\mathscr{L}_{\text{aligned}} = \text{CE}_{\text{dec}} + \boldsymbol{\lambda}_{\text{enc}} (\Sigma_{\text{CE}} + \boldsymbol{\lambda}_{\text{MSE}} \text{MSE})$$

Final Effect Estimation

Regular Encoder (No alignment) — $R^2 = 0.41$

$$\sigma(\mathbf{h}_x, \mathbf{h}_y) = 0.01\sigma(\mathbf{a}_x, \mathbf{a}_y) + 0.46\mathbb{I}_{\text{syl}}(\mathbf{p}_x, \mathbf{p}_y) + 0.09\mathbb{I}_{\text{ini}}(\mathbf{p}_x, \mathbf{p}_y) + 0.10\mathbb{I}_{\text{fin}}(\mathbf{p}_x, \mathbf{p}_y) + 0.06\mathbb{I}_{\text{tone}}(\mathbf{p}_x, \mathbf{p}_y) + 0.21J(\mathbf{w}_x, \mathbf{w}_y) + 0.003\sigma(\mathbf{v}_x, \mathbf{v}_y) - 0.08$$

Phonetic Aware Encoder (with alignment) — $R^2 = 0.66$

$$\sigma(\mathbf{h}_x, \mathbf{h}_y) = 0.42\sigma(\mathbf{a}_x, \mathbf{a}_y) + 0.2\mathbb{I}_{\text{syl}}(\mathbf{p}_x, \mathbf{p}_y) + 0.12\mathbb{I}_{\text{ini}}(\mathbf{p}_x, \mathbf{p}_y) + 0.11\mathbb{I}_{\text{fin}}(\mathbf{p}_x, \mathbf{p}_y) + 0.09\mathbb{I}_{\text{tone}}(\mathbf{p}_x, \mathbf{p}_y) + 0.05J(\mathbf{w}_x, \mathbf{w}_y) - 0.003\sigma(\mathbf{v}_x, \mathbf{v}_y) - 0.05$$

$\sigma(\cdot,\cdot)$ — cosine similarity

$\mathbb{I}_{\text{trait}}(\cdot,\cdot)$ — pinyins share a trait

$J(\cdot,\cdot)$ — Jaccard Similarity

$\mathbf{h}$ — hidden state vector

$\mathbf{w}$ — wubi tokens set

$\mathbf{p}$ — pinyin decomposition

$\mathbf{v}$ — semantic embedding (BERT)

$\mathbf{a}$ — acoustic vector (pinyin embedding)

Conclusions

Phonetic alignment produces sufficient effect towards expected contextual embeddings for characters

Dimensionality reduction resulted in sufficient improvements in recovering the similarity sources

Wubi tokenization acts as a grouping basis for characters